

Class 11 : Structural Bioinformatic pt.2

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Alpha Fold Data Base(AFDB)

The EBI maintains the largest database of AlphaFold structure prediction models at: <https://alphafold.ebi.ac.uk>

From last class (before halloween) we saw that the PDB has 244,290 (Oct,2025)

The total number of protein sequences in UniProtKB is 199,579,901

Key Point: there is a tiny fraction of sequence space that has structure coverage (0.12%)

244290 / 199579901 * 100

[1] 0.1224021

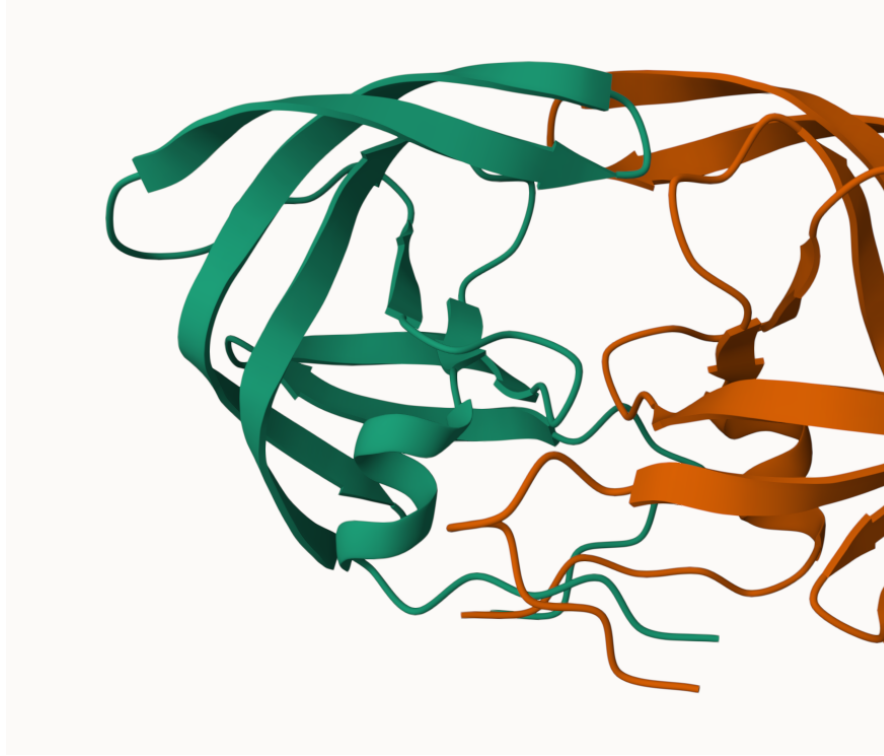
AFDB is attempting to address this gap....

There are two “Quality Scores” from AlphaFold one for residues (i.e. each amino acids) called **pLDDT** score. The other **PAE(Predicted Aligned Error)** score measures the confidence in the relative position of two residues.

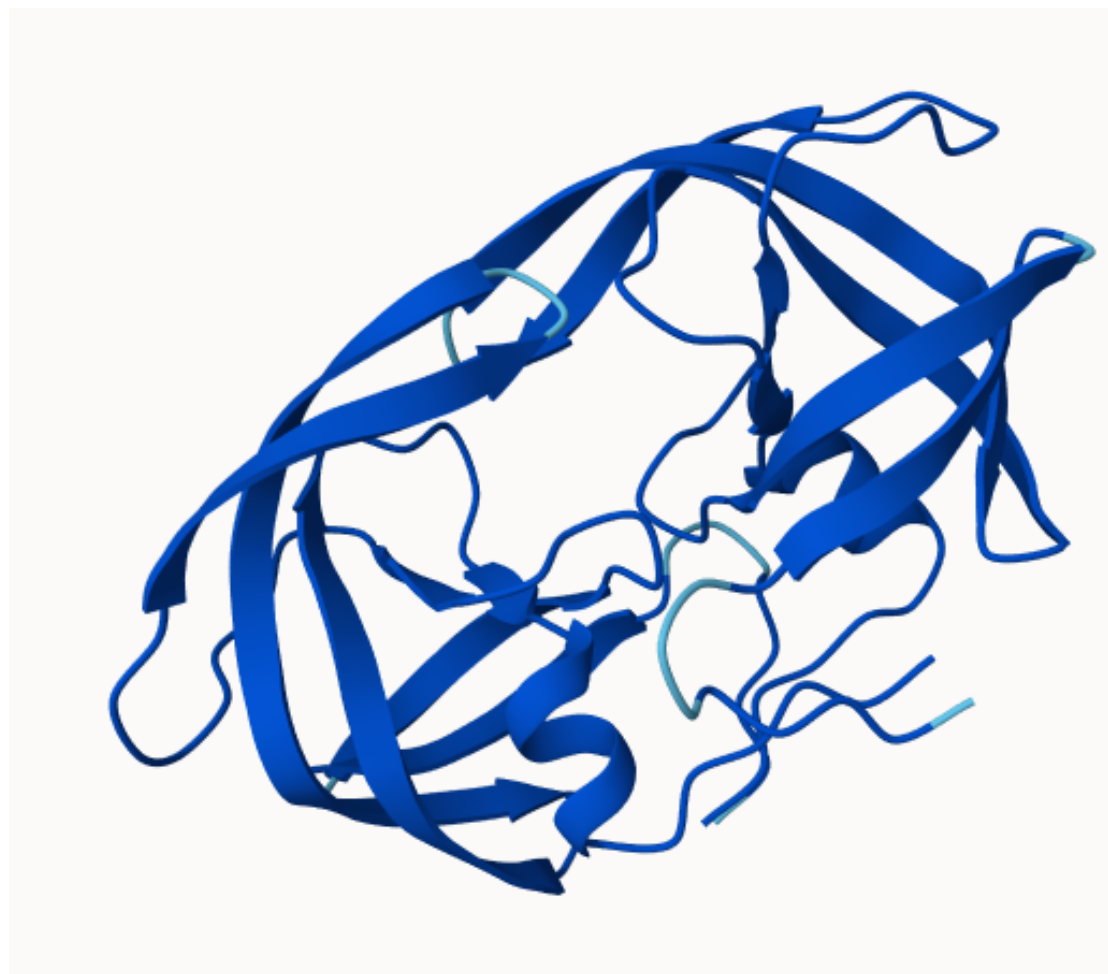
Generating your own structure predictions

Figure of 5 generated HIV-PR Models





and the top ranked model colored by chain



pLDDT score for model 1
and model 5



Custom analysis of resulting models in R

Read key results files into R. The first thing I need to know is what my results directory/folder is called (i.e. its name is different for every AlphaFold run/job)

```
results_dir <- "HIVPR_dimer_23119/"  
  
pdb_files <- list.files(path=results_dir,  
                        pattern="*.pdb",
```

```

full.names = TRUE)

# Print our PDB file names
basename(pdb_files)

```

```

[1] "HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"

```

```

library(bio3d)
m1 <- read.pdb(pdb_files[1])
m1

```

Call: read.pdb(file = pdb_files[1])

```

Total Models#: 1
Total Atoms#: 1514, XYZs#: 4542 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 0 (residues: 0)
Non-protein/nucleic resid values: [ none ]

```

```

Protein sequence:
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF

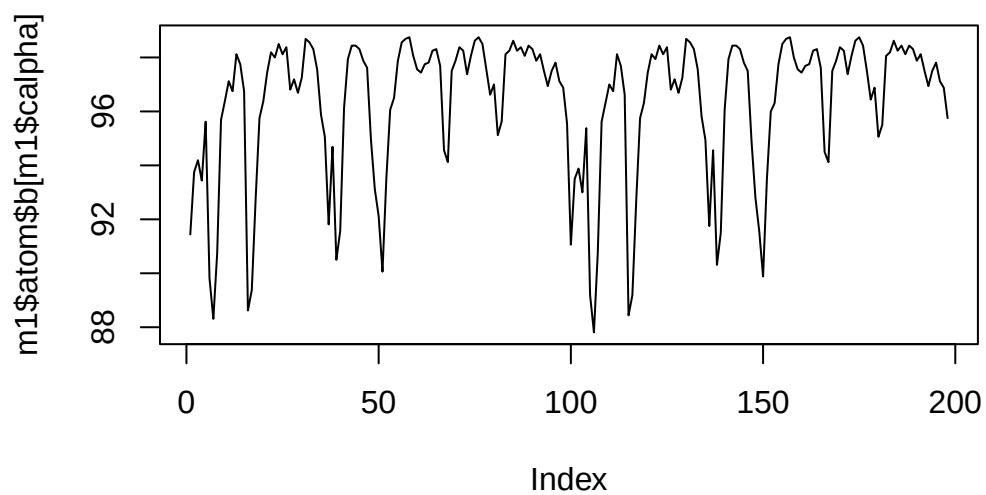
```

+ attr: atom, xyz, calpha, call

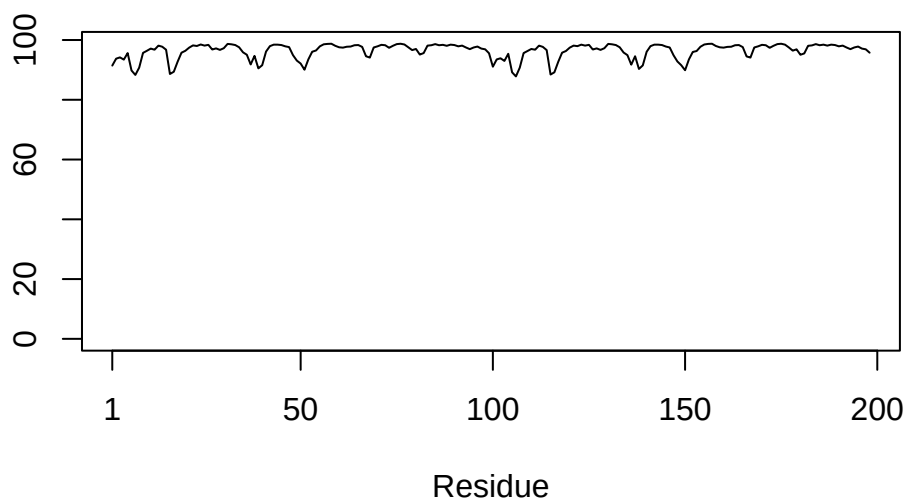
```

library(bio3d)
plot(m1$atom$b[m1$calpha],typ ="l")

```



```
plot.bio3d(m1$atom$b[m1$alpha],typ ="l")
```



##Residue conseration form alignment file

Find the large AlphaFold alignment file

```
aln_file <- list.files(path=results_dir,  
                      pattern=".a3m$",  
                      full.names = TRUE)  
aln_file
```

```
[1] "HIVPR_dimer_23119//HIVPR_dimer_23119.a3m"
```

Read this in the R

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"  
[2] " ** Duplicated sequence id's: 101 **"
```

How many sequences are in this alignment

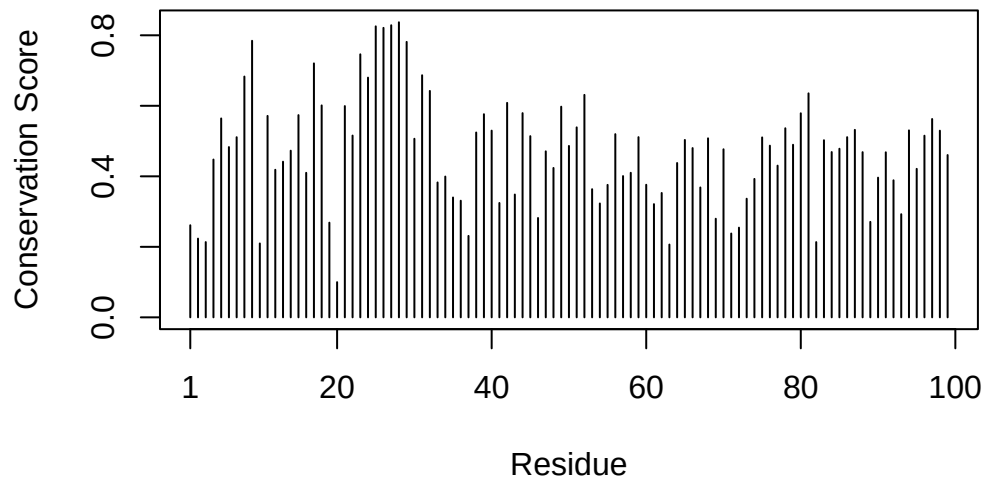
```
dim(aln$ali)
```

```
[1] 5397  132
```

We can score residue conservation in the alignment with the `conserv()` function.

```
sim <- conserv(aln)
```

```
plotb3(sim[1:99],  
       ylab="Conservation Score")
```



```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```